

THE CHINESE UNIVERSITY OF HONG KONG
Department of Statistics

STAT3003: Survey Methods
5. Systematic Sampling - Exercises

Systematic Sampling

Sometimes, performing simple random sampling is difficult, for example when population size (N) is unknown or list of sampling units is missing.

Systematic sampling with list

To obtain a systematic sample, first we choose a sample size n , then choose k . Denote N as the total number of sampling units. If N/n is an integer, let $k = N/n$; otherwise, let k be the largest integer smaller than N/n . Then find a random integer r between 1 and k , which determines the sample to be the units numbered $r, r + k, r + 2k, \dots, r + (n - 1)k$.

General definition

A sample obtained by randomly selecting one element from the first k elements in the frame and every k element thereafter is called a 1-in- k systematic sample with a random start.

Advantages:

1. Systematic sampling might be easier to conduct in the field than a simple random sample
2. If the population has a pattern (increasing or decreasing order but not periodic or cyclical order), systematic sampling may offer greater information per unit cost than SRS (smaller variance)

N	number of primary units (clusters) in the population
n	number of primary units (clusters) selected in the sample
M_i	number of secondary units (elements) in the i^{th} primary unit (cluster)
M	total number of secondary units (elements) in the population (= $\sum_{i=1}^N M_i$)
Y_{ij}	the j^{th} observation in the i^{th} primary unit
Y_i	total of observations in the i^{th} primary unit (cluster) (= $\sum_{j=1}^{M_i} Y_{ij}$)

Estimation of τ (assume the secondary units are randomly ordered)

The formulas below describe the estimation of the population total, τ , under the model where systematic sampling is treated as a form of two-stage cluster sampling with one primary unit selected.

$$\hat{t}_{sys} = NY_1 \quad Var(\hat{t}_{sys}) = \frac{M - M_1}{M - 1} \frac{M^2}{M_1} \sigma^2$$
$$\widehat{Var}(\hat{t}_{sys}) = M^2 \left(1 - \frac{M_1}{M}\right) \frac{1}{M_1} \hat{\sigma}^2 = M^2 \left(1 - \frac{M_1}{M}\right) \frac{1}{M_1 - 1} \sum_{j=1}^{M_1} (Y_{1j} - \bar{Y}_1)^2$$

The above formula is when we treat our sample of M_1 secondary units in the single primary unit chosen as if it were a SRS of size $m - 1$ from $M = \sum_{i=1}^N M_i$ secondary units in the population, which means we are using formula from **Simple Random Sampling** $Var(\hat{\tau}) = \frac{N^2}{n} (1 - \frac{n}{N}) \hat{\sigma}^2$

If the secondary units (elements) in the population are not randomly ordered, it can be shown that $Var(\hat{t}_{sys})$ depends on ICC (within-sample correlation), the correlation between pairs of secondary units (elements) in the primary unit (cluster) chosen.

Cluster sampling gives the variance of the estimator is

$$Var(\hat{\tau}) = N^2 \frac{N - n}{N - 1} \frac{\bar{M}}{n} \sigma^2 (1 + (\bar{M} - 1) ICC) \quad (1)$$

$$\approx Var(\hat{SRS})(1 + (\bar{M} - 1) ICC)$$

Sign and magnitude of ICC will affect the performance of the systematic sampling estimator, where ICC relates to patterns in the population. Notice that if ICC is positive, then the elements within the sample are similar with respect to the characteristic being measured, then the variance for estimator will expect to be higher than that in SRS. In contrast, if ICC is negative, the systematic sampling will expect to be more accurate than SRS. Consider three kinds of population:

Population	Description	Performance of $Var(\hat{t}_{sys})$
Random	Elements of population are in random order	Accurate
Ordered	Elements of population have values that trend up or down when listed (ICC is negative) as latter elements are numerically more different from former elements	Overestimate
Periodic	Elements of population have values that cycle in a regular pattern when listed (ICC is positive)	Underestimate

Repeated Systematic Sampling

Since the estimation of variance of systematic sampling relies on that the systematic sample should be close to simple random sample. We would like to have alternative method to estimate the variance of systematic sampling estimator.

Idea: Holding the (overall) sample size constant, variance should be more or less the same for any sub-samples, multiple samples (with smaller sub-sample size) can be used to provide a more representative (comparing to a single sample) estimate of the variance.

n_s : number of repeated systematic samples (usually $n_s \geq 10$)

Let the single sample we also considered be a 1-in- k (e.g. 1-in-5) systematic sample with a random start

Let the repeated samples be 1-in- k' systematic samples with random starts (where, $k' = kn_s$)

Result: n_s systematic samples (each contains n/n_s elements) (e.g. 10 times 1-in-50)

(Note: Repeated systematic sampling is essentially equivalent to cluster sampling)

Then population total τ can be estimated by:

$$\hat{t} = \frac{N}{n} \sum_{i=1}^{n_s} Y_i \quad Var(\hat{t}) = N(N - n) \frac{1}{n} \sigma_c^2,$$

where σ_c^2 is the sample variance of the cluster totals.

Exercise

1. (Adapted from Exercise 7.6 in Scheaffer et al. (2012)) The quality control section of an industrial firm uses systematic sampling to estimate the average amount of fill in 12-ounce cans coming off an assembly line. The data in Table 1 represents a 1-in-50 systematic sample of the production in one day. Estimate μ and place a 95% confident bound on the error of estimation. Assume $N = 1800$.

Amount of fill (oz)					
12.00	11.97	12.01	12.03	12.01	11.80
11.91	11.98	12.03	11.98	12.00	11.83
11.87	12.01	11.98	11.87	11.90	11.88
12.05	11.87	11.91	11.93	11.94	11.89
11.75	11.93	11.95	11.97	11.93	12.05
11.85	11.98	11.87	12.05	12.02	12.04

Table 1: Can Survey Data

2. (Adapted from Exercise 7.7 in Scheaffer et al. (2012)) Use the data in the question above to estimate μ to within 0.03 units. How many sample we need to collect?
3. (Adapted from Exercise 7.18 in Scheaffer et al. (2012)) Table 2 shows the number of births (in thousands) and the birth rate (in births per thousand of population) in the United States for a systematic sample of years between 1950 and 1990.

Year	1950	1955	1960	1965	1970	1975	1980	1985	1990
Births	3632	4097	4258	3760	3731	3144	3612	3761	4158
Rate	24.1	25.0	23.7	19.4	18.4	14.6	15.9	15.8	16.7

Table 2: Birth Survey Data

- (a) Estimate the total number of births during this 41-year period. Find an appropriate estimate of the variance.
 - (b) Estimate the mean birth rate during this period and find an appropriate estimator of the variance. Do you think this mean would be a good predictor of the birth rat for 1995? Explain
4. (Adapted from Exercise 7.22 in Scheaffer et al. (2012)) An auditor is confronted with a long list of accounts receivable for a firm. She must verify the amounts on 10% of these accounts and estimate the average difference between the audited and book values.
 - (a) Suppose the accounts are arranged chronologically, with the older accounts tending to have smaller values. Should you choose a systematic or a simple random sampling design to select the sample?
 - (b) Suppose the accounts are arranged randomly. Should you choose a systematic or a SRS design to select the sample?
 - (c) Suppose the accounts are grouped by department and then listed chronologically within departments. The older accounts again tend to have smaller values. Should you choose a systematic or a SRS design to select the sample?

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